

Homework 1

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Important: For this homework and all subsequent homeworks, whenever you are asked to analyze a particular algorithm or to design an algorithm, your solution must include a **clear English description** of your thought process. Remember, your audience is a human, not a computer, and we want to understand the process you used to arrive at the solution. This way, we can provide appropriate feedback.

1. Suppose that there are n distinct types of baseball cards, and you may purchase a box which contains a baseball card whose type is drawn independently of other boxes and uniformly at random. That is, the card in the box is equally likely to be of any of the n types. In this problem, you will compute the expected number of boxes you will need to buy in order to collect all n card types.

- (a) What is the probability that a box you buy contains a new type of card, given that you already have $k - 1$ distinct types? (Here, we are looking for an expression of the probability which depends on the two parameters n and k).

Clear English Description (a): *The thought behind our solution is directly associated with the total number of cards available n . For each pull, there are n possible distinct cards (draws). For a "success", our numerator would need to be one of the cards that we don't have, which would be $(k-1)$, but that gives the number of cards that we already have. In this case, we are looking for the cards that we don't have (Clear English Description!), which would be n (total cards) - $k-1$ (cards we do have).*

Legend: *Let A be the event box you buy contains a new type of card, given that you already have $k - 1$ distinct types*

$$Pr[A] = \frac{n - (k - 1)}{n}$$

- (b) What is the probability that it takes exactly ℓ boxes before you find a card type that you do not already have, given that you already have $k - 1$ distinct card types? (Once more, we are looking for an expression of the probability which depends on n , k , and ℓ).

Clear English Description (b): *The thought behind our solution is calculating the probability for the $\ell - 1$ failures and multiply that by the probability of success for one draw to get the probability of drawing exactly ℓ boxes before finding a card type I do not have already have.*

Legend: Let B be the event where it takes l boxes before you find a card type that you do not already have, given that you already have $k - 1$ distinct card types.

$$(1 - \Pr[A])^{l-1} * \Pr[A] = \left(\frac{(k-1)}{n}\right)^{l-1} * \frac{n - (k-1)}{n}$$

- (c) Let $\mathbf{X}_k$ denote the random variable which encodes the number of additional boxes that you need to buy in order to obtain the k -th new type of card, given that you already have $k - 1$ distinct types of cards. Use part (b) in order to express the expectation of $\mathbf{X}_k$ as a function of n and k ? You may use the fact that for any $0 < \gamma < 1$,

Clear English Description (c): Here we are tasked with deriving the expectation of the geometric mean described by Problem B. X_k follows the geometric distribution with probability of success $\Pr[A]$. In a geometric distribution, the expectation is $\frac{1}{\Pr[\text{Success}]}$. In this case, $\Pr[\text{Success}] = \Pr[A]$

$$\sum_{\ell=1}^{\infty} \ell \cdot \gamma^{\ell-1} = \frac{1}{(1-\gamma)^2}.$$

$$E[X_k] = \frac{1}{\Pr[A]}$$

$$E[X_k] = \frac{n}{n - (k - 1)}$$

- (d) Let $\mathbf{T}$ denote the total number of boxes that you need to buy in order to collect all n distinct card types. Express $\mathbf{T}$ in terms of the random variables $\mathbf{X}_k$, for potentially various values of k .

Clear English Description (d): Since T describes the total number of boxes we need to buy in order to collect all n distinct card types, let's segment this step by step: For card number 1, we will need the RV X_1 to describe the number of additional boxes to get the 1st new type of card. For card number 2, we will need the RV X_2 to describe the number of additional boxes to get the 2nd new type of card. ... For card number n , we will need the RV X_n to describe the number of additional boxes to get the n -th new type of card. Since T denotes the TOTAL number of boxes, T is the sum of all the random variables X_k such that $k = 1, 2, \dots, n$.

$$T = X_1 + X_2 + X_3 + \dots + X_n = \sum_{k=1}^n X_k$$

- (e) Compute $\mathbf{E}[T]$ using linearity of expectation, and express the final answer in terms of n and the harmonic sum $H_{n-1} = \sum_{j=1}^{n-1} 1/j$. In order to simplify the resulting expression, it is sometimes useful to consider a finite summation in the reverse order, namely,

Clear English Description (e) : *Using the expectation found from part (c), we are now going to iterate over all cases and sum them up. Next, we will substitute the summation using the linearity of expectation. Then once we have substituted using linearity of expectation, we will plug the real values and simplify to demonstrate the harmonic sum.*

$$\sum_{k=a}^b f_k = \sum_{k=b}^a f_k.$$

$$E[T] = \sum_{k=1}^n \frac{n}{n-k+1}$$

$$E[T] = E[X_1] + E[X_2] \dots + E[X_n] = \sum_{k=1}^n E[X_k]$$

$$E[T] = n * \left(\frac{1}{1} + \frac{1}{2} \dots \frac{1}{n} \right) = n * H_n$$

2. There is a flour manufacturing facility where a machine is designed to fill bags with 500 grams of flour. However, the machine is not perfect, and there are slight variations in the filling process. Each time the machine fills a bag, the number of grams in the bag can be expressed by a random variable $\mathbf{W}$ whose expectation $\mathbf{E}[\mathbf{W}] = \mu$ is 500 and whose standard deviation σ is 10 (equivalently, the variance $\sigma^2 = \mathbf{E}[(\mathbf{W} - \mu)^2]$ is 100). Each time the machine fills a bag, the corresponding random variable is independent and identically distributed.

- (a) Using Chebyshev's inequality, give an upper bound on the probability that a particular bag is filled with fewer than 475 grams or more than 525 grams. (We are expecting a solution which gives a specific number).

Clear English Description (a): *We begin with Chebyshev's inequality definition. Then we plug in the values given: variance = 100, range is 475 to 525, finally we solve for the probability that the event occurs outside of this range.*

$$\textbf{Chebyshev's Inequality: } P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

$$P(|X - 500| \geq 25) \leq \frac{1}{k^2}$$

$$k\sigma = 25$$

$$\text{If the variance is 100, then } \sqrt{100} = \sigma = 10$$

$$k \cdot 10 = 25$$

$$\frac{25}{10} = k = 2.5$$

$$P(\text{outside } [475, 525]) \leq \frac{1}{2.5^2} = 0.16$$

- (b) Suppose the machine fills 100 bags of flour. Use the fact the number of grams per bag is independent and identically distributed, as well as Chebyshev's inequality, in order to upper bound the probability that the total number of grams in all 100 bags is fewer than 49000 or more than 51000. Recall that the variance of a sum of independent random variables is the sum of the variances; namely, for k draws $\mathbf{W}_1, \dots, \mathbf{W}_k$,

$$\mathbf{E} \left[\left(\sum_{\ell=1}^k (\mathbf{W}_\ell - \mu) \right)^2 \right] = \sum_{\ell=1}^k \mathbf{E}[(\mathbf{W}_\ell - \mu)^2].$$

Clear English Description (b): *First, we need to find the expectation for the sum of the hundred bags of flour. That's simply the sum of the original expectation times one hundred. Then, we need to calculate the standard deviation based off the upper bounds and lower bounds difference. Given the calculated $k\sigma$ value, we can substitute in the known values and determine K .*

$$E[S]_{i=1}^{100} W_i = 500 \cdot 100 = 50,000$$

$$Var[S] = 100_i + 100_{i+1} \dots 100_{100} = 100 \cdot 100 = 10,000$$

$$\sigma = \sqrt{10000}$$

$$k\sigma = 51,000 - 50,000$$

$$k \cdot 100 = 1000$$

$$k = 10$$

$$P(\text{outside } [49,000, 51000]) \leq \frac{1}{10^2} = 0.01$$

3. Typically, professional sport competitions are used to identify the best player among a group. However, the International Tennis Federation (the international governing body of tennis, or ITF) wants to design a tournament—but not to identify the best—rather, the ITF wants to identify a *median* player. Formally, if there are n total players with an (unknown) ordering of skill among them, a median player has at least $n/2$ higher-skilled players and at least $n/2$ lower-skilled players.¹

The problem is that organizing a tournament with every player is too expensive, and the ITF will organize a small tournament with the hopes of identifying an ϵ -approximate median player: a player where at least $(1 - \epsilon)n/2$ players are higher-skilled and at least $(1 - \epsilon)n/2$ are lower-skilled.

¹Intuitively, this is the “middle-rank” player (in the case n is odd, there is a unique such player, and if n is even, there are two such players).

- (a) The ITF will select an odd number t between 1 and n , and choose a uniformly random set of t players to participate in the tournament (t is odd so that there is a unique median player). Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ denote n indicator random variables which are set to 0 or 1, and

$$\mathbf{X}_i = \mathbf{1} \{\text{player } i \text{ participates in the tournament}\} \in \{0, 1\}.$$

Without loss of generality, player 1 is the lowest-skill, followed by player 2, and so on until player n , who is the highest-skilled player—the indices are solely for the analysis, since the ITF does not know the ranking! For any range $a < b$, compute

$$\mu(a, b) = \mathbf{E} \left[\sum_{i=a}^b \mathbf{X}_i \right],$$

as a function of n , t , and a, b .

Clear English Description: *To compute μ we start by applying the linearity of expectation to the equation. Then we calculate $\mathbf{E}[X_i]$. Since t out of n players are selected uniformly to participate in the tournament and since $X_i = 1$ denotes player i participating in a tournament, we can deduce that $\mathbf{E}[X_i]$ is t divided by n . All that's left is to multiply that result by $(b-a+1)$, which we got from distributing the sum, to get the final results*

According to the linearity of expectations, we have:

$$\mu(a, b) = \mathbf{E} \left[\sum_{i=a}^b \mathbf{X}_i \right] = \sum_{i=a}^b \mathbf{E}[X_i]$$

We know that

$$\mathbf{E}[X_i] = \frac{t}{n}$$

because

$$\mathbf{E}[X_i] = \text{Pr}[X_i = 1]$$

So:

$$\mu(a, b) = \sum_{i=a}^b \frac{t}{n} = (b - a + 1) \frac{t}{n}$$

- (b) Even though the random variables $\mathbf{X}_1, \dots, \mathbf{X}_n$ are *not* independent, $\text{Var} \left(\sum_{i=a}^b \mathbf{X}_i \right)$ can be upper bounded by the corresponding independent version. Namely, suppose $\mathbf{X}'_1, \dots, \mathbf{X}'_n \in \{0, 1\}$ were independent and identically distributed random variables which are set to 1 with probability t/n , then $\text{Var} \left(\sum_{i=a}^b \mathbf{X}_i \right) \leq \text{Var} \left(\sum_{i=a}^b \mathbf{X}'_i \right)$. Compute $\text{Var} \left(\sum_{i=a}^b \mathbf{X}'_i \right)$ as a function of n , t , a and b in order to obtain an upper bound on $\text{Var} \left(\sum_{i=a}^b \mathbf{X}_i \right)$.

Clear English Description:

Because:

$\mathbf{X}'_1, \dots, \mathbf{X}'_n \in \{0, 1\}$ are independent:

$$\text{Var}\left(\sum_{i=a}^b \mathbf{X}'_i\right) = \text{Var}(X_a) + \text{Var}(X_{a+1}) + \dots + \text{Var}(X_b)$$

Why? Because the covariance between the variables are 0 (given they are independent from one another). So we can confidently say that the variance of a sum of the variables is the sum of the variance of its variables.

And we know that

$$\text{Var}(X_a) = \text{Var}(X_{a+1}) = \dots = \text{Var}(X_b) = \text{Var}(X_i)$$

Therefore:

$$\text{Var}\left(\sum_{i=a}^b \mathbf{X}'_i\right) = (b - a + 1) * \text{Var}(X_i) = (b - a + 1) * p * (1 - p)$$

with p being the probability of success.

Hence:

$$\text{Var}\left(\sum_{i=a}^b \mathbf{X}'_i\right) = (b - a + 1) * \text{Var}(X_i) = (b - a + 1) * \frac{t}{n} * \frac{n - t}{n} = (b - a + 1) \left[\frac{t(n - t)}{n^2} \right]$$

In conclusion:

$$\text{Var}\left(\sum_{i=a}^b \mathbf{X}_i\right) \leq (b - a + 1) \left[\frac{t(n - t)}{n^2} \right]$$

Given parts (a) and (b), we are now equipped to analyze how the median player in the tournament can be used to deduce an ϵ -approximate median player of the full tournament.

- (c) Once the tournament on the selected player is run, the ITF learns the relative ranking of the t participating players. Hence, the ITF selects the median player among those who participated in the tournament (in other words, the player so that at least $t/2$ were lower-skilled, and at least $t/2$ were higher-skilled). Let $\mathbf{m} \in \{1, \dots, n\}$ be the random variable which selects the median player among those which participated in the tournament. Justify the following three expressions:

- $\mathbf{X}_{\mathbf{m}} = 1$.

- $X_1 + X_2 + \cdots + X_m \geq t/2$.
- $X_m + X_{m+2} + \cdots + X_n \geq t/2$.

$$X_m = 1$$

Justification:

This is intuitive because $X_i = 1$ if player i participates in the tournament. By design, the median player has to have participated in the tournament. Therefore: $X_m = 1$

$$X_1 + X_2 + \cdots + X_m \geq t/2$$

Justification:

$X_i = 1$ if player i participates in the tournament. Since m represents the median player, the number of players that participate in the tournament that are less skilled than player m is $t/2$ and all of their indices must be lower than m . We know that the players are ordered with 1 being the least skilled, and n being the most skilled. Therefore the number of players spanning from player 1 until player $m-1$ must at either equal or larger than the number of selected players that participated in the tournament that are less skilled than m , which in itself equals $t/2$. And when we add m into the mix, we have that the number of players spanning from player 1 until player m is larger or equal to $t/2$.

$$\text{So: } X_1 + X_2 + \cdots + X_m \geq t/2.$$

$$X_m + X_{m+2} + \cdots + X_n \geq t/2.$$

Justification:

$X_i = 1$ if player i participates in the tournament. Since m represents the median player, the number of players that participate in the tournament that are more skilled than player m is $t/2$ and all of their indices must be higher than m . We know that the players are ordered with 1 being the least skilled, and n being the most skilled. Therefore the number of players spanning from player $m+1$ until player n must at either equal or larger than the number of selected players that participated in the tournament that are more skilled than m , which in itself equals $t/2$. And when we add m into the mix, we have that the number of players spanning from player m until player n is larger or equal to $t/2$.

$$\text{So: } X_m + X_{m+2} + \cdots + X_n \geq t/2.$$

- (d) Let $\ell_0 = (1-\epsilon)n/2$ and $\ell_1 = (1+\epsilon)n/2$, and notice that the ITF will be incorrect in identifying an ϵ -approximate median player whenever the selected player $\mathbf{m}$ lies *outside* the interval (ℓ_0, ℓ_1) —this is because the players in that interval are all of the ϵ -approximate median players. Justify the following expression:

$$\Pr[\mathbf{m} \notin (\ell_0, \ell_1)] \leq \Pr\left[\sum_{i=1}^{\ell_0} X_i > \frac{t}{2}\right] + \Pr\left[\sum_{i=\ell_1}^n X_i > \frac{t}{2}\right].$$

Let $m \leq \ell_0$.

We know from part C: $\mathbf{X}_1 + \mathbf{X}_2 + \cdots + \mathbf{X}_m \geq t/2$

In other words: $\sum_{i=1}^{\ell_0} \mathbf{X}_i \geq \sum_{i=1}^m \mathbf{X}_i \geq t/2$ because once again we are taking $m \leq \ell_0$ and the sum of the variables that span over some indices will be higher than a sum of the variables that span over a subset of those indices. In this case, the indices 1,...,m are a subset of 1,.. ℓ_0 with $m \leq \ell_0$

Therefore: $Pr[m \leq \ell_0] \leq Pr[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}]$

Similarly:

Let $m \geq \ell_1$.

We know from part C: $\mathbf{X}_m + \mathbf{X}_{m+1} + \cdots + \mathbf{X}_n \geq t/2$

In other words: $\sum_{i=\ell_1}^n \mathbf{X}_i \geq \sum_{i=m}^n \mathbf{X}_i \geq t/2$ because once again we are taking $m \leq \ell_0$ and the sum of the variables that span over some indices will be higher than a sum of the variables that span over a subset of those indices. In this case, the indices m,...,n are a subset of ℓ_1 ,..n with $m \geq \ell_1$

Therefore: $Pr[m \geq \ell_1] \leq Pr[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}]$

So $Pr[\mathbf{m} \notin (\ell_0, \ell_1)] \leq Pr[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}] + Pr[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}]$

Because the probability of the union of two events has to be less than or equal to the sum of its individual probabilities

- (e) Use Chebyshev's inequality (as well as the quantities computed in parts (a) and (b)) in order to upper bound the expressions

$$Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}\right] \quad \text{and} \quad Pr\left[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}\right]$$

as a function of n, t and ϵ . (Hint: you may find it useful to subtract the expectation from both sides of the inequality!)

$$Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}\right] = Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - E[X] > \frac{t}{2} - E[X]\right]$$

we already established from question (a) that:

$$\mu(a, b) = \sum_{i=a}^b \frac{t}{n} = (b - a + 1) \frac{t}{n}$$

Therefore: $\mu(1, \ell_0) = \frac{\ell_0 t}{n}$

Hence:

$$\begin{aligned} \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - E[X] > \frac{t}{2} - E[X]\right] &= \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t}{2} - \ell_0 \frac{t}{n}\right] \\ &= \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t}{2} - \frac{(1-\epsilon)t}{2}\right] = \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t\epsilon}{2}\right] \end{aligned}$$

According to Chebyshev's inequality:

$$\Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t\epsilon}{2}\right] \leq \frac{\text{Var}(\sum_{i=1}^{\ell_0} \mathbf{X}_i)}{(\frac{t\epsilon}{2})^2}$$

$$\begin{aligned} \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t\epsilon}{2}\right] &\leq \frac{\frac{(1-\epsilon)t(n-t)}{2n}}{\frac{t^2 \epsilon^2}{4}} \\ \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i - \mu(1, \ell_0) > \frac{t\epsilon}{2}\right] &\leq \frac{2(1-\epsilon)(n-t)}{nt\epsilon^2} \end{aligned}$$

$$\text{In other words: } \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}\right] \leq \frac{2(1-\epsilon)(n-t)}{nt\epsilon^2}$$

and that is the upper bound we are looking for

Looking at the second term

$\Pr\left[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}\right]$ is symmetric to what we did above because ℓ_0 and ℓ_1 are symmetric, and the number of terms between 0 and ℓ_0 equals to the number of terms between ℓ_1 and n with the median m being the center while each individual has similar expectation of $E[X_i] = t/n$. Therefore, we can apply the same logic as above to conclude that:

$$\Pr\left[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}\right] \leq \frac{2(1-\epsilon)(n-t)}{nt\epsilon^2}$$

- (f) Conclude that, by setting $t = O(1/\epsilon^2)$, the ITF will identify an ϵ -approximate median with probability at least 0.9.

$$\text{We know that (d): } \Pr[\mathbf{m} \notin (\ell_0, \ell_1)] \leq \Pr\left[\sum_{i=1}^{\ell_0} \mathbf{X}_i > \frac{t}{2}\right] + \Pr\left[\sum_{i=\ell_1}^n \mathbf{X}_i > \frac{t}{2}\right]$$

$$\text{Combining (d) and (e): } \Pr[\mathbf{m} \notin (\ell_0, \ell_1)] \leq \frac{4(1-\epsilon)(n-t)}{nt\epsilon^2}$$

This pretty much represents the probability of failure.

So if we want a probability of success of at least 0.9. The probability of failure has to be at most 0.1.

$$\text{Therefore: } \frac{4(1-\epsilon)(n-t)}{nt\epsilon^2} \leq 0.1$$

Solving for t :

$$t \geq \frac{4(1-\epsilon)n}{4(1-\epsilon)+0.1\epsilon^2n}$$

Now the goal is to try to make this $\frac{4(1-\epsilon)n}{4(1-\epsilon)+0.1\epsilon^2n}$ look like $O(1/\epsilon^2)$ which is what is required by the problem.

$$\frac{4(1-\epsilon)n}{4(1-\epsilon)+0.1\epsilon^2n} \leq \frac{4(1-\epsilon)n}{0.1\epsilon^2n} \leq \frac{4n}{0.1\epsilon^2n}$$

So if we take any t that is larger than $\frac{4n}{0.1\epsilon^2} = 40/\epsilon^2$. In other words, any t such that $t = O(1/\epsilon^2)$, the ITF will identify an ϵ -approximate median with probability at least 0.9